NUCLEAR FISSION

The phenomenon of division of a heavy nuclear like $_{92}U^{235}$, $_{92}U^{238}$, & that of all trans uranic elements into two nearly equal parts by projectile like slow or fast moving neutron α particle, proton, deuteron γ ray photon etc. is called **Nuclear fission**.

The product nuclei and product particles have smaller rest mass compared to the total rest mass of the reactants. This difference in mass is called mass defect. An equivalent amount of energy corresponding to this mass defect is produced according to Einstain's mass energy equivalence. ($E = \Delta mc^2$. E = energy, Δm = mass defect, C = vel of light in **vacuum**.

E x of a typical fission reaction. ; →

$$_0n^1 + {}_{92}U^{235} \longrightarrow \{{}_{92}U^{236}\}^* \text{ highly unstable compound nucleus}$$

$$\longrightarrow \underbrace{x + y}_{\text{fission fragments}} + 2 \text{ or } 3 \text{ neutrons} + \underbrace{Q}_{\text{energy released in fission}}$$

(i) The fission products x, y are highly radioactive. They decay by emitting β particle.

(ii) In each fission on an average 2.5 neutrons and 200 MeV of energy are released.

Schematic fission reaction

$$\underbrace{_0n^1}_{\text{slow neutron}} + {}_{92}U^{235} \longrightarrow \{{}_{92}U^{236}\}^* \text{ (highly unstable compound nucleus)}$$

$$\longrightarrow \underbrace{X + Y}_{\text{fission fragments}} + 2 \text{ or } 3 \text{ neutrons} + \underbrace{Q \text{ (energy)}}_{200 \text{ MeV}}$$

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NUCLEAR FUSION

If two or more light nuclei fuse or combine together to form a relatively heavier nucleus then there is a consequent decrease in nuclear rest mass which results in ~~positive~~ positive Q value [Q = energy released in nuclear reaction] according to Einstein's mass-energy equivalence [$E = \Delta mc^2$]. This type of nuclear reaction is known as nuclear fusion.

Ex. : The source of Sun's huge energy is a typical fusion reaction which occurs in four steps in which 4 protons combine together to form a helium nucleus, two positrons and 24.7 MeV of energy. This fusion cycle is called proton - proton cycle.

In symbol,

$$4\,{}_1H^1 \text{ ——— } {}_2He^4 + 2\,{}_{+1}e^0 + 24.7 \text{ MeV.}$$

Condition for fusion :

(i) Very high temperature (~10^8K) is required to initiate a fusion. High temperature provides sufficient K.E. to the reacting nuclei. So that they can overcome their mutual electrostatic repulsion & combine.

(ii) High density of matter is required so that frequent collisions can take place between the reacting nuclei.

Note: Very high temperature (~10^8K) is required to initiate a nuclear fusion, which is not available on the surface of earth.

So on earth a nuclear fission must precede a nuclear fusion.

The energy released in a nuclear fission generates heat and temp of the order of 10^8K. At this favourable temp a nuclear fusion can take place.

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